

DIFFERENTIAL EQUATIONS

LECTURE NOTES

WEEK XIII

BOUNDARY VALUE PROBLEM

TWO-POINT BOUNDARY VALUE PROBLEM (BVP)

→ The problem given by

$$y'' + p(x)y' + q(x)y = p(x) \quad x \in (a, b)$$

$$y(a) = y_0 \quad y(b) = y_1$$

general values.

where a and b are end points (boundary points)

$$\begin{array}{cc} a & b \\ y(a) = y_0 & y(b) = y_1 \end{array}$$

On the other hand In $\mathbb{R}^n$

$$y'' + p(x)y' + q(x)y = p(x) \quad x \geq x_0$$

$$y(x_0) = y_0 \quad y'(x_0) = y'_0$$

→ For the solution of B.V.P, find a function $y(x) = \phi(x)$ that satisfies the D.E in $a < x < b$ and the boundary conditions at $x = a$ and $x = b$

Ex $y'' - y = 0 \quad y(0) = 1 \quad y(1) = 0$ B.V.P.

Char eq. $\Delta(r) = r^2 - 1 = 0 \Rightarrow r_{1,2} = \pm 1$

$y_g = c_1 e^x + c_2 e^{-x}$ Find c_1, c_2 if exist by using B.Cs.

$y(0) = 1 = c_1 + c_2$

$y(1) = 0 = c_1 e + c_2 \frac{1}{e}$

$\begin{bmatrix} 1 & 1 \\ e & 1/e \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow$ has unique solution

$\det = \frac{1}{e} \neq 0$ where $c_1 = \frac{e^2 - 1}{e^2 - 1}$

(1) $c_2 = \frac{e}{e^2 - 1}$

The B.V.P has a unique solution

$$y(t) = \frac{e^2 - 1 - e}{e^2 - 1} e^t + \frac{e}{e^2 - 1} e^{-t}$$

Remark In general a B.V.P may have (i) a-unique solution.
(ii) infinitely many solutions (iii) no solutions.

Definition Consider

$$\begin{aligned} y'' + p(x)y' + q(x)y &= f(x) \\ y(a) &= y_0 \quad y(b) = y_1 \end{aligned}$$

If $f(x) = 0$ and $y_0 = y_1 = 0$, then the B.V.P is called **homogeneous**. Otherwise, **non-homogeneous**.

Now consider the homogeneous B.V.P

$$\text{B.V.P } \begin{cases} y'' + \lambda y = 0 & 0 < x < \pi \\ y(0) = 0 \quad y(\pi) = 0 \end{cases} \quad \text{where } \lambda \text{ is a parameter}$$

Note that $y(x) = \phi(x) = 0$ is always a solution for this

B.V.P **trivial (zero) solution**.

Question For which values of λ , does the B.V.P have non-trivial (non-zero) solutions?

Defn The values λ for which non-trivial solutions occur are called **eigenvalues** and these non-trivial solutions are called **eigenfunctions**.

Ex 1) $\lambda = 3$ $y'' + 3y = 0 \Rightarrow r^2 + 3 = 0 \Rightarrow r = \pm \sqrt{3}i$

$$y(x) = C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x$$

B.C. $y(0) = 0 = C_1 + 0 \Rightarrow C_1 = 0$

$$y(\pi) = 0 = C_1 \cos \sqrt{3}\pi + C_2 \sin \sqrt{3}\pi = C_2 \sin \sqrt{3}\pi \Rightarrow C_2 = 0$$

$y = 0$ is the only solution. No non-zero solution.

So, $\lambda = 3$ is not an eigenvalue of B.V.P

2) $\lambda = 1$ $y'' + y = 0 \Rightarrow r^2 + 1 = 0 \Rightarrow r = \pm i$

$$y(x) = C_1 \cos x + C_2 \sin x$$

$$y(0) = 0 = C_1 \quad y(\pi) = 0 = C_1 \cos \pi + C_2 \sin \pi \Rightarrow C_1 = 0, C_2 \text{ is arbitrary}$$

So, we have the solution $y(x) = C_2 \sin x$ where C_2 is arbitrary. We have infinitely many non-trivial solutions. Then, $\lambda = 1$ is an eigenvalue with eigenfunction $y(x) = C_2 \sin x$

Now, let us turn to the question of finding other eigenvalues and eigenfunctions of the problem.

$$y'' + \lambda y = 0$$

$$y(0) = y(\pi) = 0$$

1) $\lambda > 0$

$$\lambda = \mu^2 \quad \mu \neq 0$$

$$y'' + \lambda y = y'' + \mu^2 y = 0 \quad r^2 + \mu^2 = 0 \Rightarrow r^2 = -\mu^2 \quad r_{1,2} = \pm i\mu$$

$$y(x) = C_1 \cos \mu x + C_2 \sin \mu x$$

B.C. $\rightarrow$

$$y(0) = 0 = C_1$$

$$y(\pi) = 0 = C_2 \sin \mu \pi$$

$$C_2 = 0 \text{ since } C_1 = 0 \quad y \text{ per } = 0 \text{ trivial sol}$$

$$\sin \mu \pi = 0 \Leftrightarrow \mu \pi = n\pi$$

$$\mu = n \Rightarrow \lambda = \mu^2 = n^2 \quad n \in \mathbb{Z} \setminus \{0\}$$

If $\mu = n \Rightarrow \lambda = \mu^2 = n^2$, we have non-trivial solution.

$$y(x) = C_2 \sin \mu x = C_2 \sin(n x) \quad n \in \mathbb{Z} \setminus \{0\} \text{ are eigenfunctions}$$

Corresponding to eigenvalue $\lambda = n^2, n \in \mathbb{Z} \setminus \{0\}$

① $\lambda > 0$ ② $\lambda = 0$ ③ $\lambda < 0$. (1) is in next page

② $\lambda = 0$

$$y'' = 0 \Rightarrow r^2 = 0 \Rightarrow r_{1,2} = 0$$

$$y(x) = C_1 + C_2 x \Rightarrow \text{B.C.s} \quad y(0) = 0 = C_1$$

$$y(\pi) = 0 = C_2 \pi \Rightarrow C_2 = 0$$

$y(x) \equiv 0$ ($\lambda = 0$ is NOT an eigenvalue)

③ $\lambda < 0$ Let $\lambda = -\mu^2, \mu \neq 0$

$$y'' - \mu^2 y = 0 \Rightarrow r^2 - \mu^2 = 0 \Rightarrow r = \pm \mu \text{ real distinct.}$$

$$y(x) = C_1 e^{\mu x} + C_2 e^{-\mu x}$$

B.C.s

$$y(0) = 0 = C_1 + C_2$$

$$y(\pi) = 0 = C_1 e^{\mu \pi} + C_2 e^{-\mu \pi}$$

$$\begin{bmatrix} 1 & 1 \\ e^{\mu \pi} & e^{-\mu \pi} \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\det A = e^{-\mu \pi} - e^{\mu \pi} \neq 0 \text{ for any } \mu.$$

So, the homogeneous system has only zero solution. That is

$C_1 = C_2 = 0$ so B.V.P has only trivial (zero) solution $y(x) = 0$

So $\lambda < 0$ is NOT eigenvalue

Remark In general for the B.V.P.

$$y'' + \lambda y = 0$$

$$y(0) = 0 \quad y(L) = 0$$



The nontrivial solution exists if and only if $\lambda > 0, \lambda = \mu^2$.

$$y(x) = C_1 \cos \mu x + C_2 \sin \mu x$$

$$y(0) = 0 = C_1$$

$$y(L) = 0 = C_2 \sin \mu L$$

$$C_2 = 0 \text{ since } C_1 = 0 \Rightarrow y(x) = 0$$

no nontrivial solution

$$\sin \mu L = 0 \Leftrightarrow \mu L = n\pi \Rightarrow \mu = \frac{n\pi}{L}$$

$n = 1, 2, 3, \dots$

$$\lambda = \mu^2 = \frac{n^2 \pi^2}{L^2} \text{ are eigenvalues}$$

with eigenfunctions

$$y(x) = C_2 \sin \frac{n\pi}{L} x.$$

Exm Find eigenvalues and eigenfunctions of the B.V.P

$$\begin{cases} y'' + \lambda y = 0 \\ y(0) = 0, y'(\pi) = 0. \end{cases}$$

① $\lambda > 0$: $\lambda = \mu^2, \mu \neq 0$

$$y'' + \mu^2 y = 0 \Rightarrow r_{1,2} = \pm i\mu$$

$$y(x) = c_1 \cos \mu x + c_2 \sin \mu x \quad y'(x) = -\mu c_1 \sin \mu x + \mu c_2 \cos \mu x$$

B.C.s $y(0) = 0 = c_1$

$$y'(\pi) = 0 = -\mu c_1 \sin \mu \pi + \mu c_2 \cos \mu \pi$$

$$c_2 \mu \cos \mu \pi = 0 \Rightarrow c_2 \cos \mu \pi = 0 \quad \mu \neq 0$$

$c_2 = 0 \Rightarrow y(x) = 0$ no non-trivial soln.

$$\cos \mu \pi = 0$$

$$\Rightarrow \mu \pi = (2n+1)\frac{\pi}{2} \quad n \in \mathbb{Z}$$

Eigenvalues: $\lambda = \left(\frac{2n+1}{2}\right)^2$ eigenfunctions

$$\mu = \frac{2n+1}{2} \Rightarrow \lambda = \mu^2 = \left(\frac{2n+1}{2}\right)^2$$

$$y(x) = c_2 \sin \mu x$$

$$y(x) = c_2 \sin\left(\frac{2n+1}{2} x\right)$$

② $\lambda = 0$ $y'' = 0 \Rightarrow r_{1,2} = 0$

$$y(x) = c_1 + c_2 x \Rightarrow y' = c_2$$

B.C.s $y(0) = 0 = c_1$

$$y'(\pi) = 0 = c_2 \Rightarrow y(x) \equiv 0 \text{ no non-trivial soln.}$$

so $\lambda = 0$ is

Not an eigenvalue

③ $\lambda < 0$ $\lambda = -\mu^2, \mu \neq 0 \quad y'' - \mu^2 y = 0 \Rightarrow r_{1,2} = \pm \mu$

$$y(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}$$

$$y'(x) = c_1 \mu e^{\mu x} - c_2 \mu e^{-\mu x}$$

B.C.s $y(0) = 0 = c_1 + c_2$

$$y'(\pi) = 0 = c_1 \mu e^{\mu \pi} - c_2 \mu e^{-\mu \pi}$$

$$\begin{bmatrix} 1 & 1 \\ \mu e^{\mu \pi} & -\mu e^{-\mu \pi} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\det A = -\mu (e^{-\mu \pi} + e^{\mu \pi}) \neq 0$$

System has a unique trivial solution

$y(x) \equiv 0$ no non-trivial solution $\lambda < 0$ is not eigenvalue.

FOURIER SERIES

Definition A function $f(x)$ is periodic with period T if.

$$f(x) = f(x+T) \quad \forall x$$

Ex $\cos x, \sin x$ are periodic with period $T = 2\pi$

$\tan x, \cot x$ are periodic with period π

$$\cos(\alpha x) \rightarrow \cos(\alpha(x+T)) = \cos(\alpha x + \underbrace{\alpha T}_{2\pi}) \Rightarrow \alpha T = 2\pi \quad T = \frac{2\pi}{\alpha}$$

$$\sin(\alpha x) \rightarrow T = 2\pi/\alpha$$

Ex $\cos\left(\frac{n\pi x}{L}\right), n \in \mathbb{N}$ what is T ?

$$T = \frac{2\pi}{\alpha} = \frac{2\pi}{\frac{n\pi}{L}} = \frac{2L}{n}$$

Note that if T is a period, then $nT, n \in \mathbb{N}$ is also a period.

$$nT = n \cdot \frac{2L}{n} = 2L \text{ is also a period for } \cos \frac{n\pi}{L} x$$

So, $\cos \frac{n\pi x}{L}$ is periodic with period $T = 2L$.

Definition A series of the form

$$\frac{a_0}{2} + \sum_{m=1}^{\infty} a_m \cos \frac{m\pi x}{L} + \sum_{m=1}^{\infty} b_m \sin \frac{m\pi x}{L}$$

Fourier series
with
period $2L$

If the sum converges, it defines a function $f(x)$
which is periodic of period $2L$

Aim To express, any given function $f(x)$ which is a
periodic itself with period $2L$ as a Fourier series

Suppose,

$$f(x) = \frac{a_0}{2} + \sum_{m=1}^{\infty} a_m \cos \frac{m\pi x}{L} + \sum_{m=1}^{\infty} b_m \sin \frac{m\pi x}{L} \quad (**)$$

↓
Fourier Series of $f(x)$ and it is denoted by $S f(x)$ or $F(x)$

Find a_0, a_m, b_m

(90) Integrate (**) over $-L$ to L

$$\begin{aligned} \int_{-L}^L f(x) dx &= \int_{-L}^L \frac{a_0}{2} dx + \sum_{m=1}^{\infty} a_m \int_{-L}^L \cos \left(\frac{m\pi x}{L} \right) dx + \sum_{m=1}^{\infty} b_m \int_{-L}^L \sin \left(\frac{m\pi x}{L} \right) dx \\ &= \frac{a_0}{2} (x) \Big|_{-L}^L + \sum_{m=1}^{\infty} a_m \left(\frac{L}{m\pi} \sin \left(\frac{m\pi x}{L} \right) \right) \Big|_{-L}^L + \sum_{m=1}^{\infty} b_m \left(-\frac{L}{m\pi} \cos \left(\frac{m\pi x}{L} \right) \right) \Big|_{-L}^L \end{aligned}$$

$$\begin{aligned} \int_{-L}^L f(x) dx &= L a_0 + \sum_{m=1}^{\infty} a_m \frac{L}{m\pi} (\sin m\pi - \sin(-m\pi)) \\ &\quad + \sum_{m=1}^{\infty} -\frac{b_m L}{m\pi} (\cos(m\pi) - \cos(-m\pi)) \end{aligned}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx$$

Find a_m and b_m Note that to find a_m and b_m we'll

Use the orthogonality of $\cos \frac{m\pi x}{L}$ and $\sin \frac{m\pi x}{L}$.

Definition The functions $f(x)$ and $g(x)$ are called orthogonal on $a \leq x \leq b$ if their inner product

$$\langle f(x), g(x) \rangle = \int_a^b f(x) g(x) dx = 0$$

Remark For any positive integers m and n :

$$(a) \int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \end{cases}$$

$$(b) \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \end{cases}$$

$$(c) \int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = 0 \quad \forall m \text{ and } n \Rightarrow \sin \frac{m\pi x}{L}, \cos \frac{n\pi x}{L} \text{ are orthogonal over } [-L, L].$$

Proof (a) Remember $\cos(ax) \cos(bx) = \frac{1}{2} [\cos(a+b)x + \cos(a-b)x]$

$$\text{if } m \neq n \quad \int_{-L}^L \cos \frac{m\pi x}{L} \cos \frac{n\pi x}{L} dx = \frac{1}{2} \int_{-L}^L \cos(m+n) \frac{\pi x}{L} dx + \frac{1}{2} \int_{-L}^L \cos(m-n) \frac{\pi x}{L} dx$$

$$= \frac{1}{2} \frac{L}{(m+n)\pi} \sin(m+n) \frac{\pi x}{L} \Big|_{-L}^L + \frac{1}{2} \frac{L}{(m-n)\pi} \sin(m-n) \frac{\pi x}{L} \Big|_{-L}^L$$

$$= \frac{1}{2} \frac{L}{(m+n)\pi} \left[\underbrace{\sin(m+n)\pi}_0 + \underbrace{\sin(m+n)\pi}_0 \right]$$

$$+ \frac{1}{2} \frac{L}{(m-n)\pi} \left[\underbrace{\sin(m-n)\pi}_0 + \underbrace{\sin(m-n)\pi}_0 \right] = 0$$

If $m=n$

$$\int_{-L}^L \cos\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \int_{-L}^L \cos^2\left(\frac{n\pi x}{L}\right) dx$$

$$= \int_{-L}^L \frac{\cos\left(\frac{2n\pi}{L}x\right) + 1}{2} dx = \frac{1}{2} \left[\frac{L}{2n\pi} \sin\left(\frac{2n\pi x}{L}\right) + x \right]_{-L}^L$$

$$= \frac{1}{2} \cdot 2L = L$$

Part (a) and Part (c) can be shown in a similar way.

Now find a_m Multiply (*) with $\cos\left(\frac{n\pi x}{L}\right)$ and integrate over $-L$ to L

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \int_{-L}^L \frac{a_0}{2} \cos\left(\frac{n\pi x}{L}\right) dx + \sum_{m=1}^{\infty} a_m \underbrace{\int_{-L}^L \cos\frac{m\pi x}{L} \cos\frac{n\pi x}{L} dx}_{= \begin{cases} 0 & \text{if } m \neq n \\ L & \text{if } m = n \end{cases}}$$

$$+ \sum_{m=1}^{\infty} b_m \underbrace{\int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx}_{= 0 \text{ since sine and cosine orthogonal.}}$$

$$\int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{a_0}{2} \left(\underbrace{\frac{L}{n\pi} \sin(n\pi x)}_0 \bigg|_{-L}^L + a_n \cdot L + 0 \right)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$a_m = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{m\pi x}{L}\right) dx$$

Similarly, we can obtain

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

by multiplying $(*)$ with $\sin \frac{n\pi x}{L}$ and integrating over $-L$ to L .

Thm (Fourier Series Convergence Thm)

Let f be periodic with period $2L$. Suppose that

f and f' are piecewise continuous on the interval $-L \leq x \leq L$.

and outside $f(x+2L) = f(x)$

Then, the Fourier series of $f(x)$

$$f(x) = S_f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{L}$$

where

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx, \quad a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx, \quad b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

converges to $f(x)$ at each point where f is continuous.

If f has a jump discontinuity at $x=a \in (-L, L)$ or removable discontinuity, then $\lim_{x \rightarrow a} f(x) = L \neq f(a)$.
 (left and right limits are different)

that at $x=a$, the series $S_f(x)$ converges to the value:

$$\frac{f(a+) + f(a-)}{2}$$

$$\left[\begin{array}{l} \text{Here, } f(a+) = \lim_{x \rightarrow a^+} f(x) \\ f(a-) = \lim_{x \rightarrow a^-} f(x) \end{array} \right]$$

As a result $f(x) = S_f(x)$ when f is cont at x

If f is discont at $x=a$, $S_f(x)$ converges to $\frac{f(a+) + f(a-)}{2}$

Ex 7 Find the Fourier Series (FS) for the function.

Suppose $f(x)$ is periodic function with period $T=4$

$$f(x) = \begin{cases} 0 & -2 \leq x < -1 \\ 1 & -1 \leq x \leq 0 \\ x & 0 < x \leq 1 \\ 0 & 1 < x < 2 \end{cases}$$

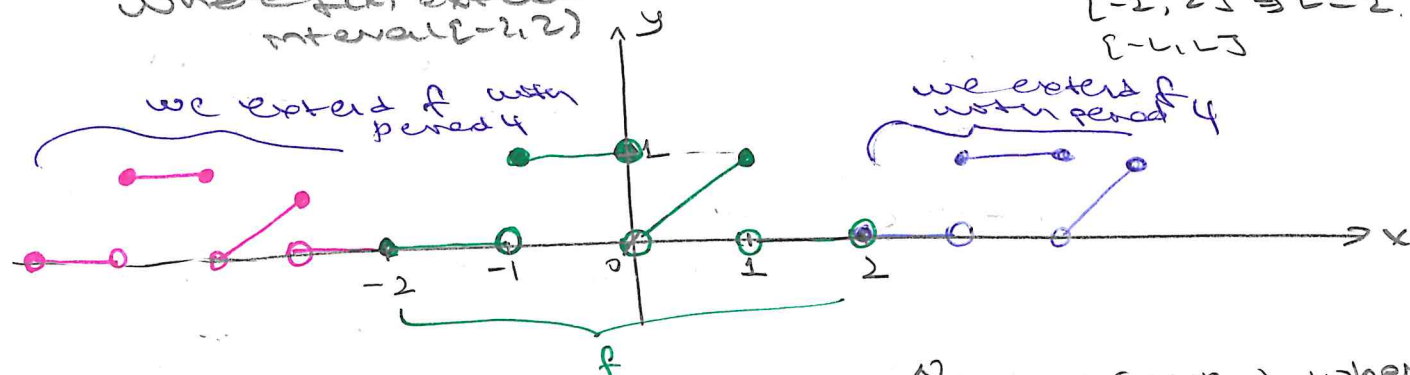
Observe that $L=2$

$$T=2L=4 \text{ that is}$$

$$f(x) = f(x+4)$$

function is defined on $[-2, 2] \Rightarrow L=2$
 $[-L, L]$

where $f(x)$ extended outside interval $[-2, 2]$



$$S_f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \text{ where}$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx = \frac{1}{2} \int_{-2}^2 f(x) dx = \frac{1}{2} \left[\int_{-2}^{-1} 0 dx + \int_{-1}^0 1 dx + \int_0^1 x dx + \int_1^2 0 dx \right]$$

$$= \frac{1}{2} \left[x \Big|_{-1}^0 + \frac{x^2}{2} \Big|_0^1 \right] = \frac{3}{4} \quad a_0 = 3/4$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} \int_{-2}^2 f(x) \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{2} \left[0 + \int_{-1}^0 1 \cdot \cos\left(\frac{n\pi x}{2}\right) dx + \int_0^1 x \cdot \cos\left(\frac{n\pi x}{2}\right) dx + 0 \right]$$

$$= \frac{1}{2} \left[\frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) \Big|_{-1}^0 + \frac{1}{2} \left[x \frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) - \frac{2}{n\pi} \int \sin\left(\frac{n\pi x}{2}\right) dx \right] \Big|_0^1 \right]$$

$$= \frac{1}{2} \frac{2}{n\pi} \sin\left(\frac{\pi n}{2}\right) + \frac{1}{2} \left[\frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) + \frac{2}{n\pi} \frac{2}{n\pi} (\cos\left(\frac{n\pi x}{2}\right)) \Big|_0^1 \right]$$

$$= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) + \frac{2}{(n\pi)^2} (\cos\left(\frac{n\pi}{2}\right) - 1) \quad n=1, 2, 3, \dots$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{1}{2} \int_{-2}^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$= \frac{1}{2} \left\{ 0 + \int_{-1}^0 \sin \frac{n\pi}{2} x dx + \int_0^1 x \sin \frac{n\pi}{2} dx + 0 \right\}$$

$$u = x \quad \sin \frac{n\pi x}{2} dx = dv$$

$$du = dx \quad -\frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) = v$$

$$= \frac{1}{2} \left[-\frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \right]_{-1}^0 \times \frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \Big|_0^1 + \frac{2}{n\pi} \int_0^1 \cos\left(\frac{n\pi x}{2}\right) dx$$

$$= -\frac{1}{n\pi} (1 - \cos \frac{n\pi}{2}) - \frac{1}{2} \left[\frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{(n\pi)^2} \sin\left(\frac{n\pi x}{2}\right) \Big|_0^1 \right]$$

$$= -\frac{1}{n\pi} + \frac{2}{(n\pi)^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\text{So, } S_f(x) = \frac{3}{6} + \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(\sin \frac{n\pi}{2} + \frac{1}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{1}{n\pi} \right) \cos\left(\frac{n\pi x}{2}\right) \\ + \sum_{n=1}^{\infty} \frac{1}{n\pi} \left(-1 + \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right) \sin\left(\frac{n\pi x}{2}\right)$$

By theorem,

$$S_f(x) \text{ converges to } \begin{cases} f(x) = 0 & -2 \leq x < -1 \\ \frac{f(-1+) + f(-1-)}{2} = \frac{1+0}{2} = \frac{1}{2} & x = -1 \\ f(x) = 1 & -1 < x < 0 \\ \frac{f(0+) + f(0-)}{2} = \frac{0+1}{2} = \frac{1}{2} & x = 0 \\ f(x) = x & 0 < x < 1 \\ \frac{f(1+) + f(1-)}{2} = \frac{0+1}{2} = \frac{1}{2} & x = 1 \\ f(x) = 0 & -1 < x \leq 2 \end{cases}$$

⑥ Calculate $S_f(-7)$ and $S_f(4\pi)$

$$S_f(-7) = S_f(-7 + 2 \cdot 4) = S_f(1) = \frac{1}{2}$$

period of $f(x)$...
and also period of $S_f(x)$

Shift the argument
in the original interval
given in problem by
adding / subtracting
 nT to argument, $n \in \mathbb{Z}$

$$S_f(4\pi) = S_f(4\pi - 3 \cdot 4) = f(4\pi - 12) = 4\pi - 12$$

$$0 < 4\pi - 12 < 1.$$

Ex 7 Find F.S for the function

$$f(x) = \begin{cases} -x & -1 < x < 0 \\ 1 & 0 < x < 1 \end{cases} \quad \text{where } f \text{ is periodically extended outside the interval } (-1, 1)$$

Observe that $L = 1$ so $T = 2L = 2$
half period

$$a_0 = \frac{1}{1} \int_{-1}^1 f(x) dx = \int_{-1}^0 (-x) dx + \int_0^1 1 dx = -\frac{x^2}{2} \Big|_{-1}^0 + x \Big|_0^1 = \frac{1}{2} + 1 = \frac{3}{2}$$

$$a_n = \frac{1}{1} \int_{-1}^1 f(x) \cos\left(\frac{n\pi x}{1}\right) dx = \int_{-1}^0 (-x) \cos(n\pi x) dx + \int_0^1 1 \cos(n\pi x) dx$$

Integration By Parts (Exercise)

$$a_n = \frac{(-1)^n - 1}{n^2 \pi^2} \quad n=1, 2, \dots$$

$$b_n = \frac{1}{1} \int_{-1}^1 f(x) \sin\left(\frac{n\pi x}{1}\right) dx = \int_{-1}^0 (-x) \sin(n\pi x) dx + \int_0^1 \sin(n\pi x) dx$$

Int. By Parts

$$= \frac{1}{n\pi} \left(1 - 2 \cos(n\pi) \right)$$

$$b_n = \frac{1}{n\pi} \left(1 - 2(-1)^n \right) \quad n=1, 2, \dots$$

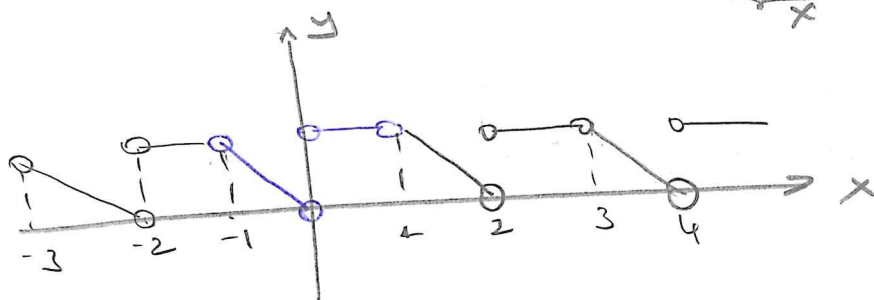
Then,

$$S_f(x) = \frac{3/2}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n - 1}{n^2 \pi^2} \cos(n\pi x) + \sum_{n=1}^{\infty} \frac{1}{n\pi} (1 - 2(-1)^n) \sin(n\pi x)$$

$$(b) S_f(3) = S_f(3 - 2) = S_f(1) = \frac{f(1^+) + f(1^-)}{2} = \frac{1+1}{2} = 1.$$

$$S_f(-2\pi) = S_f(-2\pi) = S_f(-2\pi + 3.2) = f(-2\pi + b) = -(-2\pi + b) = 2\pi - b.$$

$$-1 < -2\pi + b < 0$$



Even and odd Functions

A function $f(x)$ is odd if $f(x) = -f(-x) \quad \forall x$

A function $f(x)$ is even if $f(x) = f(-x) \quad \forall x$

Remark • If $f(x), g(x)$ are odd, then $f(x) \mp g(x)$ is odd.
 $f(x) \cdot g(x)$ is even

• If $f(x), g(x)$ are even, then $f(x) \mp g(x)$ is even
 $f(x) \cdot g(x)$ is even

• If $f(x)$ is odd, $g(x)$ is even, then $f(x) \mp g(x)$ neither even nor odd
 $f(x) \cdot g(x)$ is odd.

• If $f(x)$ is odd, then $\int_{-L}^L f(x) dx = 0$.

• If $f(x)$ is even, then $\int_{-L}^L f(x) dx = 2 \int_0^L f(x) dx$

• Even functions are symmetric w.r.t. y-axis

Odd functions are symmetric w.r.t. origin.

Now, apply these facts to Fourier series

① If $f(x)$ is odd, then $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx = 0$

$$a_n = \frac{1}{L} \int_{-L}^L \underbrace{f(x)}_{\text{odd}} \underbrace{\cos\left(\frac{n\pi x}{L}\right)}_{\text{even}} dx = 0$$

$$b_n = \frac{1}{L} \int_{-L}^L \underbrace{f(x)}_{\text{odd}} \underbrace{\sin\left(\frac{n\pi x}{L}\right)}_{\text{odd}} dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

So, Fourier series of an odd function $f(x)$ will be:

$$S_f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \text{ where } b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Which is called Fourier Sine Series (F.S.S)

② If $f(x)$ is even, then $a_0 = \frac{1}{L} \int_{-L}^L f(x) dx = \frac{2}{L} \int_0^L f(x) dx$

$$a_n = \frac{1}{L} \int_{-L}^L \underbrace{f(x)}_{\text{even}} \underbrace{\cos\left(\frac{n\pi x}{L}\right)}_{\text{even}} dx = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L \underbrace{f(x)}_{\text{even}} \underbrace{\sin\left(\frac{n\pi x}{L}\right)}_{\text{odd}} dx = 0$$

Then, the Fourier series of an even function $f(x)$ is:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

where $a_0 = \frac{2}{L} \int_0^L f(x) dx$ and $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$

Which is called Fourier Cosine Series

Remark Accordingly, by extending a function $f(x)$ defined on $0 \leq x \leq L$ oddly we get series with NO COSINE terms (and constant term (evenly))
(NO SINE)

THM Let f be a piecewise function with both f and f' are piecewise continuous on $0 \leq x \leq L$. Then,

Fourier Sine Series (FSS) $\sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$ with $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$

Fourier Cosine Series (FCS) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$ with $a_0 = \frac{2}{L} \int_0^L f(x) dx$
and $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$

both converge to $f(x)$ at each point $x \in (0, L)$ where f is continuous. If f has a jump discontinuity at $a \in (0, L)$ then both series converge to $\frac{f(a+) + f(a-)}{2}$ at $x=a$

→ At the points

$x=0$ and $x=L$

{ the FSS will be zero
the FCS will converge to $f(0+)$ at $x=0$ and $f(L-)$ at $x=L$

EVEN AND ODD EXTENSIONS OF $f(x)$

Let f be given on $[0, L]$, To obtain its Fourier series we can extend it into two periodic functions

Even extension of $f(x)$ with period $2L$ | odd extension of $f(x)$ with period $2L$

$$g(x) = \begin{cases} f(x) & 0 \leq x \leq L \\ f(-x) & -L < x < 0 \end{cases}$$

$g(x)$ is even with period $2L$, so it is called even extension of $f(x)$ with period $2L$. Then, $g(x)$ is a Fourier

$$h(x) = \begin{cases} f(x) & 0 < x < L \\ 0 & x = 0, L \\ -f(-x) & -L < x < 0 \end{cases}$$

$h(x)$ is odd with period $2L$

$S_n(x)$ is Fourier-Sine series

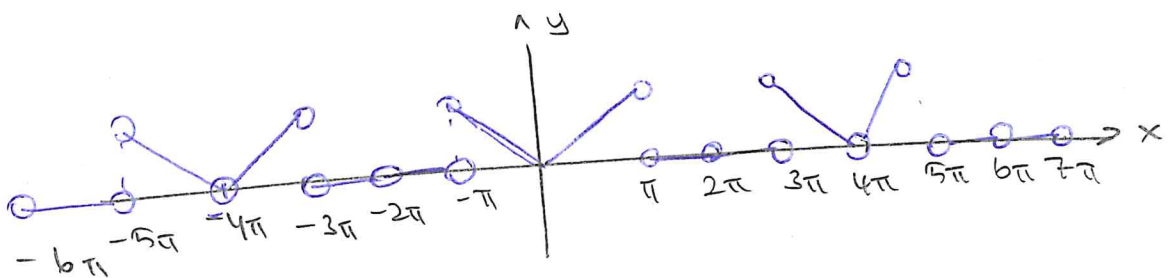
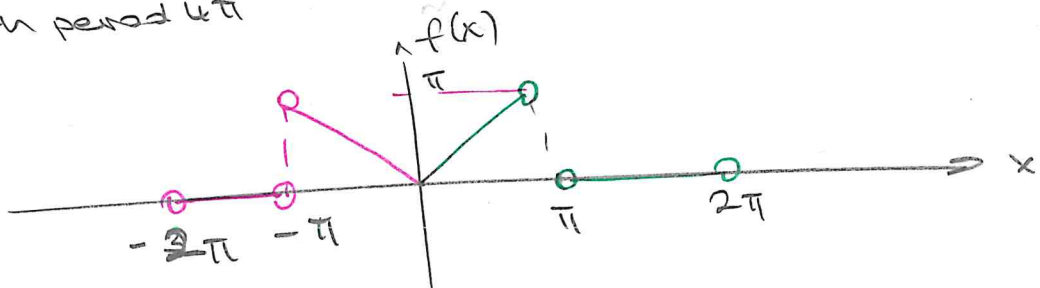
Gives even extension of f . It's Fourier series converges f on $[0, L]$ closed int. of function f . Gives odd extension of function f . It's Fourier series converges f on $(0, L)$ open interval.

Ex 7 $f(x) = \begin{cases} x & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases}$ $L = 2\pi$ $2L = 4\pi = T$

Sketch the graph of even extension of $f(x)$ and find its Fourier series for the extended function. One like f.e.m.p. note Solst

$$g(x) = \tilde{f}(x) = \begin{cases} f(x) & 0 < x < 2\pi \\ f(-x) & -2\pi < x < 0 \end{cases} = \begin{cases} x & 0 < x < \pi \\ 0 & \pi < x < 2\pi \\ -x & -\pi < x < 0 \\ 0 & -2\pi < x < -\pi \end{cases}$$

even extension of $f(x)$ with period 4π



Since f is even, we have only cosine terms in S_f^N , $b_n = 0$

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L \underbrace{\tilde{f}(x)}_{\text{even}} dx = \frac{2}{L} \int_0^L \underbrace{\tilde{f}(x)}_{=f(x)} dx = \frac{2}{L} \int_0^L f(x) dx \\ &= \frac{2}{2\pi} \int_0^{2\pi} f(x) dx \\ &= \frac{1}{\pi} \left[\int_0^{\pi} x dx + \int_{\pi}^{2\pi} 0 dx \right] \\ &= \frac{1}{\pi} \cdot \frac{\pi^2}{2} = \frac{\pi}{2} = a_0 \end{aligned}$$

$$\begin{aligned} a_n &= \frac{1}{L} \int_{-L}^L \underbrace{\tilde{f}(x)}_{\text{even}} \cos\left(\frac{n\pi x}{L}\right) dx = \frac{2}{L} \int_0^L \underbrace{\tilde{f}(x)}_{=f(x)} \cos\left(\frac{n\pi x}{L}\right) dx = \frac{2}{2\pi} \int_0^{2\pi} f(x) \cos\left(\frac{n\pi x}{2}\right) dx \\ &= \frac{1}{\pi} \left[\int_0^{\pi} x \cos\left(\frac{n}{2}x\right) dx + \int_{\pi}^{2\pi} 0 \cos\left(\frac{n}{2}x\right) dx \right] \\ &\quad \downarrow \\ &\quad \text{I.B. Parts (Exercise)} \end{aligned}$$

$$= \frac{2}{n\pi} \left[\pi \sin\left(\frac{n\pi}{2}\right) + \frac{2}{n} \left(\cos\left(\frac{n\pi}{2}\right) - 1 \right) \right]$$

$$\boxed{S_f^N(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty} \left(\frac{2}{n} \sin\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi} \left(\cos\left(\frac{n\pi}{2}\right) - 1 \right) \right) \cos\left(\frac{n}{2}x\right)}$$

$$\textcircled{b} \quad S_f^N(16) = S_f^N(16-4\pi) = \tilde{f}(16-4\pi) = f(16-4\pi) = 0$$

$$\pi < 16-4\pi < 2\pi$$

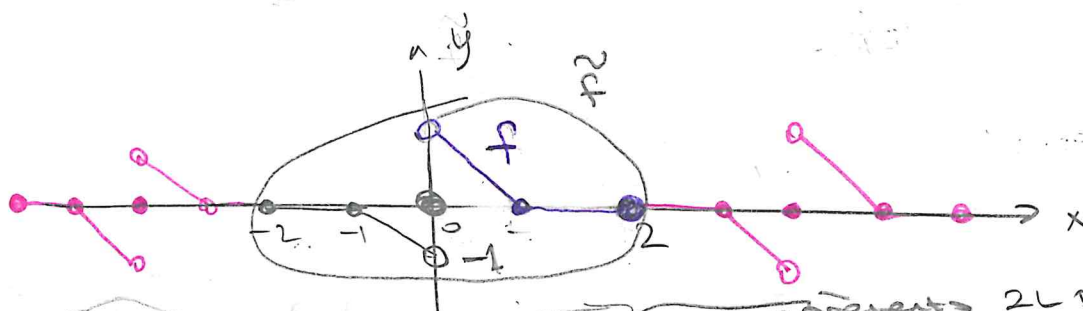
$$\begin{aligned} S_f^N(-7\pi) &= S_f^N(-7\pi + 2 \cdot 4\pi) = \tilde{f}(\pi) = \frac{\tilde{f}(\pi+) + \tilde{f}(\pi-)}{2} \\ &= \frac{0 + 1}{2} = \frac{1}{2} \end{aligned}$$

NOTE Please note that $a_0 = \frac{1}{L} \int_{-L}^L \tilde{f}(x) dx = \frac{2}{L} \int_0^L \tilde{f}(x) dx$ $x \in (0, L)$
 This is overextension of f since it is even we write equality
 Similarly $a_n = \frac{1}{L} \int_{-L}^L \tilde{f}(x) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{2}{L} \int_0^L \tilde{f}(x) \cos\left(\frac{n\pi x}{L}\right) dx$
 Since $\tilde{f}(x) = f(x)$ for $x \in (0, L)$, we make $f(x)$ equality

Ex 7 $f(x) = \begin{cases} 1-x & 0 < x < 1 \\ 0 & 1 \leq x \leq 2 \end{cases} \quad L=2, 2L=4$

① Extend $f(x)$ to all $\mathbb{R}$ or an odd function with period 4, and sketch the graph of the extension.

$$\tilde{f}(x) = \begin{cases} 1-x & 0 < x < 1 \\ 0 & 1 \leq x \leq 2 \\ 0 & x=0 \\ -f(-x) = -(1+x) & -1 < x < 0 \\ 0 & -2 \leq x \leq -1 \end{cases}$$



② Find Fourier series of the function $\tilde{f}$ around in part ①. All graphs represent $2L$ periods of $\tilde{f}$.

Since $\tilde{f}$ is odd, then Fourier series we have only sine terms, that is $a_0=0, a_n=0, \forall n \geq 1$. And

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx = \frac{2}{2} \int_0^2 f(x) \sin\left(\frac{n\pi x}{2}\right) dx \\ &= \int_0^1 (1-x) \sin\left(\frac{n\pi x}{2}\right) dx + \int_1^2 0 \cdot \sin\left(\frac{n\pi x}{2}\right) dx \\ &= \int_0^1 \sin\frac{n\pi x}{2} dx - \int_0^1 x \sin\left(\frac{n\pi x}{2}\right) dx \\ &\quad \begin{aligned} u &= x \quad dv = \sin\left(\frac{n\pi x}{2}\right) dx \\ du &= dx \quad v = -\frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \end{aligned} \\ &= -\frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \Big|_0^1 - \left[-x \frac{2}{n\pi} \cos\left(\frac{n\pi x}{2}\right) \Big|_0^1 + \frac{2}{n\pi} \int_0^1 \cos\left(\frac{n\pi x}{2}\right) dx \right] \end{aligned}$$

$$= -\frac{2}{n\pi} \left\{ \cos\left(\frac{n\pi}{2}\right) - 1 \right\} - \left[-\frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{4}{n^2\pi^2} \sin\left(\frac{n\pi x}{2}\right) \right]_{\sin\left(\frac{n\pi}{2}\right)}^1$$

$$b_n = -\frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{2}{n\pi} + \frac{2}{n\pi} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$b_n = \frac{2}{n\pi} \left[1 - \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right]$$

$$\boxed{S_n^{\sim} f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left[1 - \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right] \sin\left(\frac{n\pi}{2} x\right)}$$

$$\textcircled{c} S_n^{\sim} f(2) = \tilde{f}(2) = f(2) = 0$$

$\tilde{f}$ is cont at $x=2$.

$$S_n^{\sim} f(-4) = \tilde{f}(-4+4) = \tilde{f}(0) = \frac{\overbrace{f(0+)}^{f(0)} + \underbrace{\tilde{f}(0-)}_{\text{jump adjacent at } x=0}}{2} = \frac{1+(-1)}{2} = 0$$

Exercise $f(x) = e^x$ on $(0, \pi)$ Give the F.S for the odd exte
of $f(x) = e^x$ on the interval $(-\pi, \pi)$ and sketch its
graph.

$$\underline{\text{Sol}} \quad L = \pi, \quad 2L = 2\pi$$

$$\tilde{f}(x) = \begin{cases} e^x & 0 < x < \pi \\ 0 & x = 0, \pi \\ -f(-x) = -e^{-x} & -\pi < x < 0 \end{cases}$$

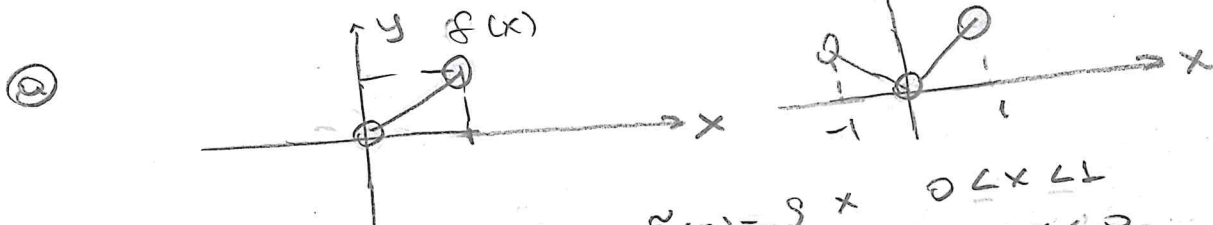
$$S_n^{\sim} f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} \left(\frac{1}{1-n^2} \right) (1 - e^{\pi(-1)^n}) \sin nx.$$

Summary example Given $f(x)$ ($f(x)$ & $f'(x)$ are piecewise continuous) on $0 < x < L$ we will extend it ~~to~~ even/odd and then make it $2L$ periodic i.e. $\tilde{f}(x+2L) = \tilde{f}(x)$ where $\tilde{f}$ is an extension.

Ex $\frac{7}{7}$ Consider $f(x) = x$ $0 < x < L$

(a) Obtain even extension, sketch the graph

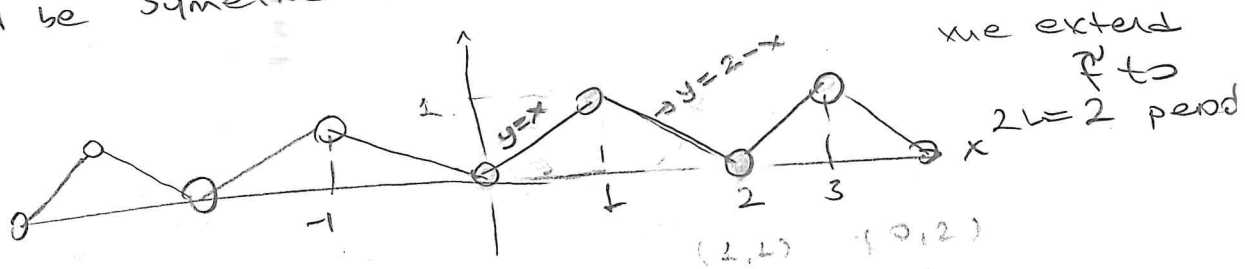
(b) Obtain odd extension, sketch the graph



$$\tilde{f}(x) = \begin{cases} f(x) & 0 < x < L \\ f(-x) & -L < x < 0 \end{cases} \quad \tilde{f}(x) = \begin{cases} x & 0 < x < L \\ -x & -L < x < 0 \end{cases}$$

even extension

we will make $\tilde{f}$ $2L$ periodic since $L=1$, then we have period 2 $\tilde{f}(x+2) = \tilde{f}(x)$ since we are even function. It will be symmetric with respect to y axis.



$$S_{\tilde{f}} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

$$L=1. \quad a_0 = \frac{1}{L} \int_{-L}^L \tilde{f}(x) dx = 2 \int_0^1 f(x) dx = 2 \cdot \frac{x^2}{2} \Big|_0^1 = 1$$

even extension

$$a_n = \frac{1}{L} \int_{-L}^L \tilde{f}(x) \cos\left(\frac{n\pi x}{L}\right) dx = 2 \int_0^1 x \cos(n\pi x) dx$$

$$= 2 \left\{ \frac{x}{n\pi} \sin(n\pi x) \Big|_0^1 - \int_0^1 \frac{1}{n\pi} \sin(n\pi x) dx \right\}$$

$$= \frac{2}{n^2 \pi^2} (\cos(n\pi) - 1) = \frac{2}{n^2 \pi^2} ((-1)^n - 1)$$

$$S_{\tilde{f}} = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi^2} ((-1)^n - 1) \cos(n\pi x)$$

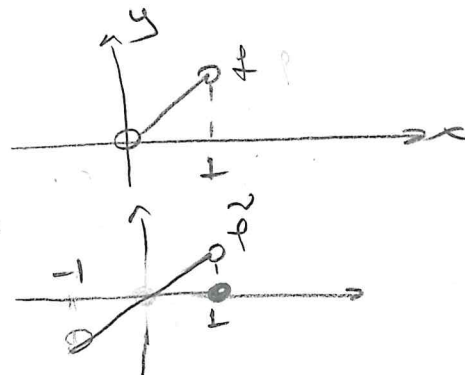
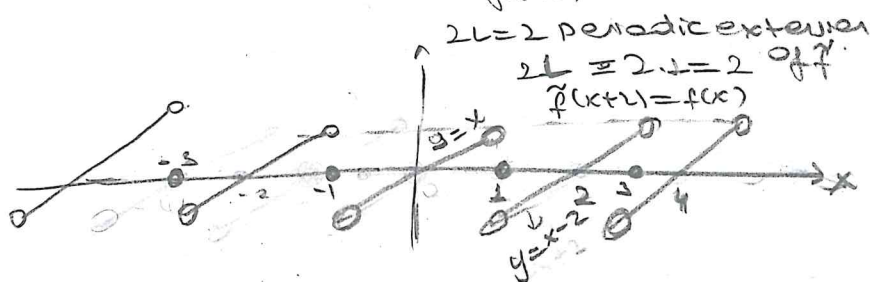
$S_{\tilde{f}}$ is the Fourier even extension of $\tilde{f}(x)$

$S_f(x)$ converges to $\tilde{f}(x)$ (where $\tilde{f}(x)$ is cont. and if $\tilde{f}(x)$ is
discont. at $x=a$, then S_f converges to $\frac{\tilde{f}(a^+) + \tilde{f}(a^-)}{2}$

so at $x=0$ S_f converges to $\frac{\tilde{f}(0^+) + \tilde{f}(0^-)}{2} = \frac{0+0}{2} = 0$
at $x=1$ S_f $\tilde{f}(1^+) = \lim_{x \rightarrow 1^+} \tilde{f}(x) = 1$ $\tilde{f}(1^-) = \lim_{x \rightarrow 1^-} \tilde{f}(x) = -1$ S_f converges to $\frac{1+(-1)}{2} = 0$

(b) Odd extension.

$$\tilde{f}(x) = \begin{cases} x=f(x) & 0 < x < 1 \\ 0 & x=0, 1 \\ -(-x) = x = -f(-x) & -1 < x < 0 \end{cases}$$



$$b_n = \frac{1}{1} \int_{-1}^1 \tilde{f}(x) \sin(n\pi x) dx = 2 \int_0^1 x \sin(n\pi x) dx = 2 \left(-x \cos(n\pi x) \Big|_0^1 + \int_0^1 \cos(n\pi x) dx \right)$$

Integrate by parts

$$x=u \quad \sin(n\pi x) dx = dv \\ dx = du \quad -\frac{\cos(n\pi x)}{n\pi} = v$$

$$= \frac{2}{n\pi} (-\cos(n\pi)) + \frac{1}{(n\pi)^2} \sin(n\pi x) \Big|_0^1$$

$$= \frac{2}{n\pi} (-1)^{n+1}$$

$$S_f(x) = \sum_{n=1}^{\infty} \frac{2}{n\pi} (-1)^{n+1} \sin(n\pi x)$$

odd extension

at $x=0$ S_f converges to $\frac{\tilde{f}(0^+) + \tilde{f}(0^-)}{2} = \frac{0+0}{2} = 0$

at $x=1$ S_f converges to $\frac{\tilde{f}(1^+) + \tilde{f}(1^-)}{2} = \frac{1+(-1)}{2} = 0$

HEAT CONDUCTION IN A ROD



$u(x,t)$: heat distribution along the rod at any time t .

Heat Conduction Problem

$$\begin{cases} u_t = \alpha^2 u_{xx} & 0 < x < L, t > 0 \\ u(x,0) = f(x) & \text{"initial condition"} \\ u(0,t) = u(L,t) = 0 & \text{"homogeneous BCs"} \end{cases}$$

α : thermal diffusivity constant

$$\left[u_t = \frac{\partial u}{\partial t}, u_x = \frac{\partial u}{\partial x}, u_{xx} = \frac{\partial^2 u}{\partial x^2} \right]$$

SEPARATION OF VARIABLES (SOV)

Look for solutions of type $u(x,t) = \bar{X}(x) \cdot T(t)$, then insert in differential equation $u_t = \alpha^2 u_{xx}$

$$u_t = \bar{X}(x) T'(t) \quad \left(T' = \frac{dT}{dt} \right)$$

$$u_x = \bar{X}'(x) T \quad u_{xx} = \bar{X}''(x) T$$

Then, $u_t = \alpha^2 u_{xx} \Rightarrow \bar{X}(x) T'(t) = \alpha^2 \bar{X}''(x) T(t)$

$$\underbrace{\frac{\bar{X}''(x)}{\bar{X}(x)}}_{\text{function of } x \text{ only}} = \underbrace{\frac{1}{\alpha^2} \frac{T'(t)}{T(t)}}_{\text{function of } t \text{ only}} = -\lambda$$

two functions are equal if the ratio is equal to a constant
let us say $-\lambda, \lambda$: any constant

$$\frac{\bar{X}''(x)}{\bar{X}(x)} = -\lambda \quad \text{and} \quad \frac{1}{\alpha^2} \frac{T'}{T} = -\lambda$$

"we obtain two ordinary DEs with $\bar{X}, T$."

$$\textcircled{1} \quad \boxed{\bar{X}'' + \lambda \bar{X} = 0}$$

$$\boxed{T' + \lambda \alpha^2 T = 0} \quad \textcircled{2}$$

For side conditions

$$u(0, t) = \bar{X}(0)T(t) = 0 \rightarrow T(t) = 0 \quad \forall t \Rightarrow u(x, t) = \bar{X}(x)T(t) = 0 \quad \text{or}$$

$$\rightarrow T(t) \neq 0 \Rightarrow \boxed{\bar{X}(0) = 0} \quad \text{only trivial solution}$$

$$u(L, t) = \bar{X}(L)T(t) = 0 \rightarrow T(t) = 0 \quad \forall t \Rightarrow u(x, t) = 0$$

$$\rightarrow T(t) \neq 0 \Rightarrow \boxed{\bar{X}(L) = 0}$$

This gives us

$$\begin{cases} \bar{X}'' + \lambda \bar{X} = 0 \\ \bar{X}(0) = \bar{X}(L) = 0 \end{cases}$$

"Two-point B.V.P. that we discuss before, find λ such that we have non-zero solutions"

Case Equation

$$\boxed{\lambda = 0} \quad r^2 = 0 \quad r = 0 \text{ repeated}$$

$$\bar{X}(x) = c_1 + c_2 x$$

$$\bar{X}(0) = c_1 = 0$$

$$\bar{X}(L) = c_1 + c_2 L \Rightarrow c_2 = 0$$

} $\bar{X}(x) = 0$ no non-zero solutions.

$$\boxed{\lambda < 0} \quad \lambda = -\mu^2 \quad \mu \neq 0$$

$$r^2 - \mu^2 = 0 \Rightarrow r = \pm \mu \text{ real distinct.}$$

$$\bar{X}(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}$$

$$\bar{X}(0) = c_1 + c_2 = 0$$

$$\bar{X}(L) = c_1 e^{\mu L} + c_2 e^{-\mu L}$$

$$\Rightarrow \begin{bmatrix} 1 & 1 \\ e^{\mu L} & e^{-\mu L} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\det \neq 0 \Rightarrow$ only trivial solution

No non-zero solution

$$\boxed{\lambda > 0} \quad \lambda = \mu^2 \quad \mu \neq 0$$

$$r^2 + \mu^2 = 0 \Rightarrow r = \pm i\mu$$

$$\bar{X}(x) = c_1 \cos \mu x + c_2 \sin \mu x$$

$$\bar{X}(0) = c_1$$

$$\bar{X}(L) = 0 = c_2 \sin \mu L \begin{cases} c_2 = 0 \\ \bar{X}(x) = 0 \\ \text{no non-zero} \end{cases}$$

$$\begin{cases} c_2 \neq 0 \\ \sin \mu L = 0 \\ \mu L = n\pi \\ L = \frac{n\pi}{\mu} \end{cases}$$

We have non-zero solution when $\lambda = \mu^2 = \left(\frac{n\pi}{L}\right)^2 \neq \frac{n^2\pi^2}{L^2}$

Eigenfunctions $\sin \mu x = \sin \frac{n\pi x}{L}$

$$X_n(x) = \sin \frac{n\pi x}{L} \quad n=1, 2, \dots \quad \lambda = \frac{n^2\pi^2}{L^2}$$

Then, the associated equation for $T(t)$ is

$$T' + \lambda \alpha^2 T = 0 \quad \text{solve for } \lambda = \frac{n^2\pi^2}{L^2}$$

"Separable" $T' + \alpha^2 \frac{n^2\pi^2}{L^2} T = 0 \Rightarrow \frac{T'}{T} = -\alpha^2 \frac{n^2\pi^2}{L^2}$

$$\ln(T) = -\alpha^2 \frac{n^2\pi^2}{L^2} t + C_1$$

$$T_n(t) = C \cdot e^{-\alpha^2 \frac{n^2\pi^2}{L^2} t} \quad n \neq 0, n=1, 2, \dots$$

So for each $n=1, 2, \dots$

$$u_n(x, t) = X_n(x) T_n(t) = \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha^2 \frac{n^2\pi^2}{L^2} t}$$

So, linear combination of each solution (for each n) is also a solution of heat conduction problem.

But, here there are infinitely many solutions. So their

linear combination is a series

$$u(x, t) = \sum_{n=1}^{\infty} c_n u_n(x, t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha^2 \frac{n^2\pi^2}{L^2} t}$$

To find coefficient c_n we use I.C $u(x,0) = f(x)$

$$u(x,0) = f(x) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) \quad \text{is a Fourier sine series}$$

so that

$$c_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

Ex 7 Find the solution to the problem

$$u_t = \alpha^2 u_{xx}$$

$$u(0,t) = u(L,t) = 0$$

$$u(x,0) = \sin\left(\frac{\pi x}{L}\right) + 3 \sin\left(\frac{5\pi x}{L}\right) \quad f(x)$$

We already found the solution of PDE above such that

$$u(x,t) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \quad \text{for } c_n \text{ apply initial condition}$$

$$\begin{aligned} u(x,0) &= \sin\left(\frac{\pi x}{L}\right) + 3 \sin\left(\frac{5\pi x}{L}\right) = \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) \\ &= c_1 \sin\left(\frac{\pi x}{L}\right) + c_2 \sin\left(\frac{2\pi x}{L}\right) + c_3 \sin\left(\frac{3\pi x}{L}\right) \\ &\quad + c_4 \sin\left(\frac{4\pi x}{L}\right) + c_5 \sin\left(\frac{5\pi x}{L}\right) + \dots \end{aligned}$$

If $c_1 = 1$ and $c_5 = 3$ and $c_n = 0$ $n = 2, 3, 4, 5, \dots$
 $n \neq 1, 5$.

$$c_n = \begin{cases} 1 & n=1 \\ 3 & n=5 \\ 0 & n \neq 1, 5 \end{cases}$$

$$\begin{aligned} \text{Then, } u(x,t) &= \sum_{n=1}^{\infty} c_n \sin\left(\frac{n\pi x}{L}\right) e^{-\alpha^2 \frac{n^2 \pi^2}{L^2} t} \\ &= \sin\left(\frac{\pi x}{L}\right) e^{-\alpha^2 \frac{\pi^2}{L^2} t} + 0 + 0 + 3 \sin\left(\frac{5\pi x}{L}\right) e^{-\alpha^2 \frac{25\pi^2}{L^2} t} + 0 \end{aligned}$$

Q2/7 You can use: OR $C_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$ to find

$$C_n = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi x}{L}\right) \left(\sin\frac{\pi x}{L} + 3 \sin\frac{5\pi x}{L} \right) dx$$

Ex Find the solution of

$$\begin{cases} u_t = \frac{1}{4} u_{xx} & 0 < x < 1, t > 0 & L=1 & \alpha^2 = \frac{1}{4} \\ u(x,0) = 5 \sin(2\pi x) - 3 \sin(5\pi x) & 0 < x < 1 \\ u(0,t) = u(1,t) = 0 \end{cases}$$

Again $u(x,t) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{L} x\right) e^{-\frac{\alpha^2 n^2 \pi^2 t}{L^2}} \quad L=1 \quad \alpha^2 = \frac{1}{4}$

$$u(x,t) = \sum_{n=1}^{\infty} C_n \sin(n\pi x) e^{-\frac{1}{4} n^2 \pi^2 t}$$

To find C_n use I.C.

$$u(x,0) = 5 \sin 2\pi x - 3 \sin 5\pi x = C_1 \sin \pi x + C_2 \sin 2\pi x + C_3 \sin 3\pi x + \dots + C_5 \sin 5\pi x + C_6 \sin 6\pi x + \dots$$

$$C_2 = 5 \text{ and } C_5 = -3, \quad C_n = 0 \quad n \neq 2, 5.$$

$$\text{Then, } u(x,t) = 5 \sin 2\pi x e^{-\frac{1}{4} 2^2 \pi^2 t} - 3 \sin(5\pi x) e^{-\frac{1}{4} 5^2 \pi^2 t}$$

Ex Solve the PDE

$$\begin{cases} u_t = u_{xx} + u & 0 < x < 2, t > 0 & L=2 \\ u(x,0) = x & \\ u_x(0,t) = 0 & u_x(2,t) = 0 \end{cases}$$

Assume $u(x,t) = X(x)T(t) \quad u_t = XT' \quad u_{xx} = X''T$

$$u_t = u_{xx} + u \Rightarrow XT' = X''T + XT$$

$$X''T + X(T - T') = 0$$

$$X''T = (T' - T)X$$

$$\frac{X''}{X} = \frac{T' - T}{T} = -\lambda$$

Func. of x -only Func. of t -only

$$(1) X'' + \lambda X = 0$$

$$u_x(0, t) = 0 \Rightarrow X'(0)T(t) = 0$$

$$T(t) = 0 \forall t$$

$$u(x, t) = X(x) \cdot T(t)$$

no non-zero solutions
x

$$T(t) \neq 0$$

$$\Rightarrow X'(0) = 0$$

similarly

$$u_x(2, t) = 0 \Rightarrow X'(2)T(t) = 0 \Rightarrow X'(2) = 0$$

$$\text{Solve the B.V.P } X'' + \lambda X = 0 \\ X'(0) = X'(2) = 0$$

$$(2) T' + (\lambda - 1)T = 0 \quad \text{"separable"}$$

$$\frac{T'}{T} = 1 - \lambda \quad \ln T = (1 - \lambda)t \quad T(t) = e^{(1 - \lambda)t}$$

$$\underline{\lambda = 0} \quad r^2 = 0 \Rightarrow r = 0$$

$$X(x) = c_1 + c_2 x \quad X'(x) = c_2 \quad X'(0) = c_2 = 0$$

$$X'(2) = 2c_2 = 0 \Rightarrow c_2 = 0$$

So $c_2 = 0$ and c_1 arbitrary $\Rightarrow X(x) = c_1$ is a non-zero solution
w.l.o.g. let $c_1 = 1$

So $X(x) = 1$ is a solution (non-zero)

$$\underline{\lambda > 0} \quad \lambda = \mu^2, \mu \neq 0$$

$$r^2 + \mu^2 = 0 \quad r_{1,2} = \pm i\mu \Rightarrow X(x) = c_1 \cos \mu x + c_2 \sin \mu x$$

$$X'(x) = -\mu c_1 \sin \mu x + \mu c_2 \cos \mu x$$

$$X'(0) = 0 = c_2 \mu \Rightarrow c_2 = 0$$

$$X'(2) = 0 = -c_1 \mu \sin 2\mu \quad \begin{cases} c_1 = 0 \Rightarrow \text{since } c_2 = 0 \Rightarrow X(x) = 0 \\ c_1 \neq 0 \Rightarrow \sin 2\mu = 0 \Rightarrow 2\mu = n\pi \quad n \neq 0 \end{cases}$$

$$\mu = \frac{n\pi}{2} \quad n = 1, 2, \dots$$

$$\underline{X_n(x) = \cos \mu x = \cos\left(\frac{n\pi x}{2}\right)}$$

$$n = 1, 2, \dots$$

(26)

$$\underline{\lambda = \mu^2 = \frac{n^2 \pi^2}{4}}$$

2.6 $\lambda = -\mu^2$ $r_{1,2} = \pm \mu$

$\bar{X}(x) = c_1 e^{\mu x} + c_2 e^{-\mu x}$

Exercise

$\Rightarrow c_1 = c_2 = 0 \Rightarrow \bar{X}(x) = 0$

no. non-zero solution

Then: we already found $T(t) = e^{(1-\lambda)t}$

$\lambda = 0$ $T_0(t) = e^t \Rightarrow u_0(x,t) = \bar{X}_0(x) \cdot T_0(t) = 1 \cdot e^t = e^t$

$\lambda = \frac{n^2 \pi^2}{4}$ $T_n(t) = e^{(1 - \frac{n^2 \pi^2}{4})t} \Rightarrow u_n(x,t) = \bar{X}_n(x) T_n(t)$
 $(n=1, 2, \dots)$
 $u_n(x,t) = \cos\left(\frac{n\pi x}{2}\right) e^{(1 - \frac{n^2 \pi^2}{4})t}$

Then, their linear combination is the solution of the given problem

$u(x,t) = \frac{c_0}{2} u_0(x,t) + \sum_{n=1}^{\infty} c_n u_n(x,t)$

$u(x,t) = \frac{c_0}{2} e^t + \sum_{n=1}^{\infty} c_n \cos\left(\frac{n\pi x}{2}\right) e^{(1 - \frac{n^2 \pi^2}{4})t}$

Find c_n by using F.C: $u(x,0) = x$

$t=0 \Rightarrow u(x,0) = x = \frac{c_0}{2} + \sum_{n=1}^{\infty} c_n \cos\left(\frac{n\pi x}{2}\right)$ "Fourier cosine Series"

Then, $c_0 = \frac{2}{L} \int_0^L x dx = \frac{x^2}{2} \Big|_0^2 = 2$

$c_n = \frac{2}{L} \int_0^L x \cos\left(\frac{n\pi x}{2}\right) dx = \int_0^2 x \cdot \cos\left(\frac{n\pi x}{2}\right) dx$

$u = x$ $\cos\left(\frac{n\pi x}{2}\right) dx = dv$
 $du = dx$ $\frac{2}{n\pi} \sin\left(\frac{n\pi x}{2}\right) = 1$

$c_n = \frac{2}{n\pi} x \cdot \sin\left(\frac{n\pi x}{2}\right) \Big|_0^2 - \frac{2}{n\pi} \int_0^2 \sin\left(\frac{n\pi x}{2}\right) dx$

$c_n = \frac{4}{n^2 \pi^2} \cos\left(\frac{n\pi x}{2}\right) \Big|_0^2 = \frac{4}{n^2 \pi^2} (\cos(n\pi) - 1)$

$u(x,t) = \frac{2}{2} e^t + \sum_{n=1}^{\infty} \frac{4}{n^2 \pi^2} ((-1)^n - 1) \cos\left(\frac{n\pi}{2} x\right) e^{(1 - \frac{n^2 \pi^2}{4})t}$

Ex Solve the problem by using separation of variables

$$u_{xx} = (1+t)u_t \quad 0 < x < 1 \quad t > 0 \quad u=1$$

$$u(0,t) = u(1,t) = 0$$

$$u(x,0) = \sin(3\pi x) + 7\sin(5\pi x) \quad 0 < x < 1$$

Assume $u(x,t) = X(x)T(t)$ $u_x = X'T$ $u_{xx} = X''T$ $u_t = XT'$

$$u_{xx} = (1+t)u_t \Rightarrow X''T = (1+t)XT'$$

$$\frac{X''}{X} = (1+t) \frac{T'}{T} = -\lambda$$

Function of x only
func of t-only

$$X'' + \lambda X = 0$$

$$u(0,t) = 0 = X(0)T(t) \Rightarrow X(0) = 0$$

$$u(1,t) = 0 = X(1)T(t) \Rightarrow X(1) = 0$$

$$(1+t)T' + \lambda T = 0$$

$$\frac{T'}{T} = -\frac{\lambda}{1+t}$$

$$\ln T = -\lambda \ln(1+t)$$

$$T = \frac{1}{(1+t)^\lambda} = (1+t)^{-\lambda}$$

$\lambda = 0$ $r=0$ $X(x) = c_1 + c_2 x$

$$\left. \begin{aligned} X(0) = 0 = c_1 \\ X(1) = 0 = c_2 \end{aligned} \right\} \begin{aligned} X &= 0 \\ \text{no non-trivial} \\ \text{solutions} \end{aligned}$$

$\lambda < 0$ $\lambda = -\mu^2, \mu \neq 0 \Rightarrow$ no non-trivial solutions (exercise)

$\lambda > 0$ $\lambda = \mu^2, \mu \neq 0 \Rightarrow \lambda = \pi^2 \mu$

$$X(x) = c_1 \cos \mu x + c_2 \sin \mu x$$

$$X(0) = 0 = c_1$$

$$X(1) = c_2 \sin \mu = 0 \quad c_2 = 0 \quad X = 0 \quad x$$

$$\sin \mu = 0 \quad \mu = n\pi \quad n \neq 0 \quad n=1, 2, \dots$$

$$X_n(x) = \sin(n\pi x) \quad \Leftarrow \quad \lambda = \mu^2 = n^2 \pi^2$$

The corresponding T when $\lambda = n^2 \pi^2$ $T_n(t) = (1+t)^{-n^2 \pi^2}$ $n=1, 2, \dots$

$$u_n(x,t) = X_n(x)T_n(t) = \sin(n\pi x) (1+t)^{-n^2 \pi^2}$$

Then, $u(x,t) = \sum_{n=1}^{\infty} c_n u_n(x,t) = \sum_{n=1}^{\infty} c_n \sin(n\pi x) (1+t)^{-n^2 \pi^2}$

Find c_n by using I.C $u(x,0) = \sin 3\pi x + 7\sin 5\pi x$

$t=0$ $u(x,0) = \sum_{n=1}^{\infty} c_n \sin(n\pi x) \cdot 1$ "Fourier sine series" (28)

$$U(x,0) = \sin 3\pi x + 7 \sin 5\pi x = \underset{\substack{\parallel \\ 0}}{C_1} \sin \pi x + \underset{\substack{\parallel \\ 0}}{C_2} \sin 2\pi x + \underset{\substack{\parallel \\ 0}}{C_3} \sin 3\pi x + \underset{\substack{\parallel \\ 0}}{C_4} \sin 4\pi x \\ + \underset{\substack{\parallel \\ 7}}{C_5} \sin 5\pi x + \underset{\substack{\parallel \\ 0}}{C_6} + \dots$$

$$\text{So } c_n = \begin{cases} 1 & n=3 \\ 7 & n=5 \\ 0 & n \neq 3, 5 \end{cases}$$

$$U(x,t) = \sum_{n=1}^{\infty} c_n \sin(n\pi x) (1+t)^{-n^2\pi^2}$$

$$U(x,t) = \underset{\substack{\parallel \\ 1}}{C_3} \sin(3\pi x) (1+t)^{-9\pi^2} + \underset{\substack{\parallel \\ 7}}{C_5} \sin(5\pi x) (1+t)^{-25\pi^2}$$

Exercise

$$2U_{xx} = U_t$$

$$0 < x < 6 \quad t > 0$$

$$U_x(0,t) = 0$$

$$U_x(6,t) = 0$$

$$U(x,0) = \pi + 3\cos\pi x - 4\cos\left(\frac{3\pi}{2}x\right) - \cos(3\pi x)$$

$$\text{Ans } T_n(t) = e^{-\frac{2n^2\pi^2}{36}t}$$

$$U_n(x,t) = \underbrace{e^{-\frac{2(n\pi)^2}{36}t}}_{T_n} \underbrace{\cos\left(\frac{n\pi}{6}\right)x}_{X_n}$$

$$\textcircled{C_0 = \pi} \quad C_6 = 3 \quad C_9 = 4 \quad C_{12} = -1 \quad C_n = 0 \quad n \neq 0, 6, 9, 12$$

$$U(x,t) = \sum_{n=0}^{\infty} c_n U_n(x,t) = \textcircled{C_0 U_0} + C_6 U_6 + \dots$$

